

BlockBase – Day 1

Mentored by FEC

Proof Of Work (Bitcoin)

▶ What is Proof of Work? (Cryptocurrency Explanation)

Proof of work (PoW) is a blockchain consensus mechanism that requires significant computing effort from a network of devices. The concept was adapted from digital tokens by Hal Finney in 2004.

Following its introduction in 2009, Bitcoin became the first widely adopted application of Finney's PoW idea (Finney was also the recipient of the first bitcoin transaction). Proof of work is also the mechanic used in many other cryptocurrencies.

KEY TAKEAWAYS

- Proof of work (PoW) is a decentralized consensus mechanism that requires network members to expend effort in solving an encryption puzzle.
- Proof of work is also called mining, in reference to receiving a reward for work done.
- Proof of work allows for secure peer-to-peer transaction processing without needing a trusted third party.
- Proof of work at scale requires vast amounts of energy, which only increases as more miners join the network.

Understanding Proof of Work

Proof of work is a concept used in some public blockchains to demonstrate that a program did the work required to propose a new block for the chain. It is commonly called a consensus mechanism because, eventually, network consensus is reached after there is proof the work was done honestly (in this case, "honestly" means there were no attempts to alter data).

Proof of work is provided by sending the information in a block through a hashing algorithm, then adjusting variable fields until a particular hexadecimal number is reached. This serves as proof that the program expended the computational effort to "hash" the block until a solution was reached.

Proof of Work and Consensus

Here's a quick rundown of the proof of work process on the Bitcoin blockchain.

First, the worker, which is called a miner, creates a temporary file (a block). If it wins the competition to solve for a winning hash, this file will be stored on the blockchain. The block has the four following fields:

- Block size
- Block header
- Transaction counter
- Transactions

The block header contains the following fields:

- Software version
- Previous block's hash
- Merkle root
- Timestamp
- Difficulty target
- Nonce

The mining program assembles this block and places the transactions it has prioritized in the transaction field. It continuously adjusts the nonce and the extra nonce and sends the information in the block through a hashing algorithm.

It repeats this process until it finds a solution, which is a value less than or equal to the difficulty target. The difficulty target is set so that a certain number of hashes per second must be attempted before a solution is found. For example, on May 17, 2024, block 843,900 had a difficulty target of 83.148T, or 83.148 trillion attempts per second per miner.

This hash provided proof to the network that the miner did the work. The block was added to the blockchain, and the network began its process of reaching consensus.

This whole process requires extensive computational power, electricity and huge funds.

Proof-of-work (PoW) fundamentals

The work(ers)

Proof-of-work blockchains are supported through a network of decentralized computers called nodes. They are tasked with two things: accepting batches of transactions from other nodes and validating (or proposing) new blocks of transactions to the network.

These nodes are also called miners because they spend computing power and resources in return for the network's underlying cryptocurrency.

The "work" in proof-of-work is the computational power nodes have to contribute in validating a new block of transactions. This power is represented by the SHA-256 cryptographic hash function (the one we read in Day 1 task 2) that sets this consensus mechanism apart from its counterparts.

An algorithm called the difficulty adjustment ensures that it will take the entire network a fixed set of time to validate new blocks of transactions. The difficulty adjustment occurs approximately every 2,016 blocks (about once every two weeks) to maintain the target block time of 10 minutes. Miners coming and going from the network on an individual basis do nothing to affect difficulty level minute to minute, or day to day.

Miners win the reward when they guess a hash that falls below the threshold provided by the network. Once a miner finds the valid block hash, it broadcasts this information to other miners who can quickly validate and add the new block to their blockchain copies. This validation process eliminates the possibility of miners including malicious transactions.

The Reward

There is an encoded rule regarding the amount paid to the miner who completes the proof-of-work. Currently, miners earn a fixed 3.125 BTC per block, plus any user transaction fees. This reward potential incentivizes miners to compete in the proof-of-work and remain honest, as any attempt to cheat the system would waste resources.

The reward amount is set to half every 210,000 blocks (approximately four years). This deflation rate is referred to as the halving cycle. Many fear that if bitcoin's price fails to keep pace, miners will lose the incentive to participate. But as miners disconnect from the network, the difficulty level drops accordingly. This balancing act makes the cost to mine bitcoin drop as well.

The economics of bitcoin mining though are not straight forward. There are many financing factors that drive miners to stay online even when they are unprofitable.

Proof Of Stake (Ethereum)

 What is Proof of Stake? How it works (Animated) + Ethereum 2.0 Upgrade!

A blockchain protocol provides traders with incentives to validate transactions by rewarding them with cryptocurrency for every correct validation. As a safeguard against fraud, proof-of-stake protocols require traders to “stake” some of their cryptocurrency as collateral, which is then locked up in a deposit. If a trader adds a transaction to the blockchain that other validators deem to be invalid, they can lose a portion of what they staked.

There's usually a lower limit to how much validators can stake. After the limit is surpassed, validators can stake as much as they want. In fact, the more a trader stakes, the more likely they are to be chosen by the algorithm. Here's a simple example to illustrate the point: let's say there's a new change to the blockchain that needs verification. Ten nodes volunteer to validate it, and they each stake one cryptocurrency for the privilege. That means that they each have an equal 10 percent chance of being awarded the work.

Let's say that one volunteer really wants to win the work. They could up the odds by staking three coins on the deal. If everyone else kept their stake at one coin, they would up their chance of winning the work to 25 percent, while everyone else's chances would go down to 8.3 percent.

In practice, it's a lot more complicated than that. That's because new transactions are grouped together in blocks, sometimes of several hundred or more. Then several blocks are chained together to create a record of all the transactions in order. Another complicating factor is that traders can enter staking pools, where groups of validators can together come up with the lower limit to become a validator. When a staking pool is awarded the work, the reward is split among the pool's members, with a slightly larger share going to the pool's owner.

The Proof Of Stake algorithm uses a pseudo-random election process to select validators from a group of nodes. The system uses a combination of factors, including staking age, an element of randomization, and the node's wealth.

Users participating in the forging process must lock a certain amount of coins into the network as their stake. The stakes' size determines the chances for a node to be selected as the next validator as mentioned above.

Advantages of Proof of Stake

Proof of Stake has many clear advantages over Proof of Work. For this reason, new blockchains almost always use Proof of Stake. Its benefits include:

Adaptability

As users' needs and blockchains change, so can Proof of Stake. It's plain to see this with the vast number of adaptations available. The mechanism is versatile and can easily fit most blockchain use cases.

Decentralization

More users are encouraged to run nodes since it's more affordable. This incentive and the randomization process also make the network more decentralized. Although staking pools exist, there is a much higher chance for an individual to forge a block under Proof of Stake successfully. Overall, this reduces the need for staking pools.

Energy efficiency

Proof of Stake is incredibly energy efficient compared to Proof of Work. The cost of participating relies on the economic cost of staking coins rather than the computational cost of solving puzzles. This mechanism leads to a significant reduction in the energy required to run the consensus mechanism.

Scalability

As Proof of Stake doesn't rely on physical machines to generate consensus, it's more scalable. There's no need for huge mining farms or sourcing large energy supplies. Adding more validators to the network is cheaper, simpler, and more accessible.

Security

Staking works as a financial motivator for the validator not to process fraudulent transactions. If the network detects a fraudulent transaction, the validator will lose a part of its stake and its right to participate in the future. So as long as the stake is higher than the reward, the validator would lose more coins than it would gain with fraudulent activity.

To effectively control the network and approve fraudulent transactions, a node would have to own a majority stake in the network, also known as the 51% attack. Depending on the value of a cryptocurrency, it can be almost impossible to gain control of the network, as you would need to acquire 51% of the circulating supply.

Disadvantages of Proof of Stake

Although Proof of Stake has many advantages compared to Proof of Work, it still has some weaknesses:

Forking

With a standard Proof of Stake mechanism, there's no disincentive for mining both sides of a fork. Under Proof of Work, mining both sides will lead to a waste of energy. With Proof of Stake, the cost is much less, meaning that people can "bet" on both sides of a fork.

Accessibility

To begin staking, you'll need a blockchain's native token supply. This requires you to purchase the token via an exchange or other method. Depending on the amount required, you may need a significant investment to begin staking effectively.

With Proof of Work, you can buy cheap mining equipment or even rent it. With this, you can join a pool and start validating and earning quickly.

51% attack

While Proof of Work is also prone to 51% attacks, they can be significantly easier with Proof of Stake. If a token's price crashes or the blockchain has a low market capitalization, it can be theoretically cheap to purchase more than 50% of the tokens and control the network.

Proof Of Stake VS Proof Of Work

▶ Proof-of-Stake (vs proof-of-work)

▶ Proof of Work vs Proof of Stake: Which is Better? (ANIMATED)

Reading Material : [Proof of Work vs. Proof of Stake: Comparing blockchain consensus](#)